

Assignment 3 – Group 5

Financial Engineering: Buy Side
Covariance Estimation Techniques & Statistical Factor Models

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Introduction

This report presents the analysis conducted for the Equity Statistical Arbitrage at Polimi Capital Management. We study covariance estimation techniques on the Euro Stoxx 50 constituents, performing monthly rebalancing (end of month) with a rolling 2-year estimation window. The analysis is organized in three parts: linear shrinkage of the covariance matrix, PCA-based statistical factor models, and the connection between the two approaches.

To execute these tasks, we utilize several auxiliary Python modules:

- `backtest.py`: Aligns portfolio weights with historical asset returns to compute realized and cumulative performance over time.
- `covariance_utilities.py`: Prepares the rolling estimation windows by filtering out assets with insufficient data and validates the covariance matrices.
- `portfolio_optimization.py`: Implements closed-form solutions to calculate the optimal weights for Minimum Variance and Mean-Variance portfolios.
- `principal_component_analysis.py`: Handles PCA decomposition, aligns eigenvectors across periods to prevent sign flips, and reconstructs PCA-denoised covariance matrices.
- `shrinkage.py`: Computes the optimal shrinkage intensity and the final shrunk covariance matrices using both Constant Correlation and Market Factor targets.

Software Environment Notice

All results in this report were produced using `pandas 3.0.1` and `numpy 2.4.3`. Earlier versions of `pandas` may produce incorrect numerical outputs due to changes in default behavior (e.g. `groupby`, `cov`, index alignment). Please ensure these versions are installed before running the code.

Part I: Covariance Estimation

Task 1a: Risk Aversion Calibration

We consider the first full estimation window (504 trading days) ending on December 31, 2014, with 46 eligible assets. The mean-variance portfolio is defined by the following optimization problem:

$$\max_w \mu^\top w - \frac{\gamma}{2} w^\top \Sigma w, \quad \text{s.t.} \quad \mathbf{1}^\top w = 1$$

where μ and Σ are the sample mean and covariance matrix, respectively.

As specified in the assignment, the strategy must target a Gross Exposure (GE) of 3. The GE

represents the sum of the absolute weights of the portfolio:

$$\text{GE}(w) = \sum_{i=1}^N |w_i| = 3$$

Since the portfolio weights w are a function of the risk aversion parameter γ , but no analytical inverse mapping exists, we solve for γ numerically. We define the objective function $f(\gamma)$ as the difference between the portfolio's GE and our target:

$$f(\gamma) = \left(\sum_{i=1}^N |w_i(\gamma)| \right) - 3$$

We use Brent's method to find the root $f(\gamma^*) = 0$. This ensures that the resulting mean-variance portfolio utilizes the exact amount of leverage (long and short positions) required by the task. The calibrated value is $\gamma = 158.10$.

We compare the Minimum Variance Portfolio (MVP) with our calibrated Mean-Variance (MV) portfolio.

Strategy	Sum of Weights	Mean Return	Volatility	Gross Exposure
Minimum-Variance	1.0	0.000468	0.006151	2.778
Mean-Variance ($\gamma = 158.10$)	1.0	0.000851	0.006345	3.000

Table 1: Comparison between GMVP and Calibrated MV Portfolio

The results show a divergence in Gross Exposure (GE) between the two strategies that reflects their distinct optimization objectives and constraints:

- **Minimum Variance Portfolio (MVP):** Pure risk minimization — the gross exposure of 2.778 is not a choice but the natural outcome of the optimization. Forcing it to 3.0 would add leverage without any return objective to justify it, increasing portfolio variance for no compensated reason.
- **Mean-Variance (MV):** Risk-adjusted return maximization. We calibrate $\gamma = 158.10$ so that $\text{GE} = 3.0$ exactly. The extra exposure relative to MVP is compensated: it allows the optimizer to move toward higher-expected-return assets, moving the portfolio along the efficient frontier.

Tasks 1b–1c: Shrinkage Covariance Estimation

To mitigate the noise in the sample covariance matrix S , we implement a linear shrinkage framework. The shrunk estimator $\hat{\Sigma}$ is a weighted average of the sample covariance and a structured target F :

$$\hat{\Sigma} = \delta F + (1 - \delta)S, \quad \delta \in [0, 1]$$

The shrinkage intensity δ is calibrated to minimize the estimation error (Frobenius norm). We implement two different targets:

Constant Correlation Target (CC)

The **Constant Correlation** target assumes that all pairs of assets share the same average correlation $\bar{\rho}$, while keeping their individual sample variances.

$$f_{ii} = s_{ii}, \quad f_{ij} = \bar{\rho} \sqrt{s_{ii} s_{jj}}$$

By pulling all correlations toward the global mean $\bar{\rho}$, this method effectively neutralizes extreme, "noisy" correlation values that often arise by pure statistical chance in the sample matrix.

Single-Factor Target (SF)

The **Single-Factor** target assumes that asset returns are driven by a single common source of risk: the market (represented by an equal-weighted index). The target matrix is structured as:

$$F_{SF} = \sigma_{mkt}^2 \beta \beta^\top + \Delta$$

where β are the sensitivities (betas) of each asset to the market and Δ contains the residual variances. This model is highly stable because it filters out all pairwise relationships that cannot be explained by the general market movement.

Task 1d: Portfolio Construction

For each of the three covariance estimators (sample, constant correlation shrinkage, market factor shrinkage), we compute the minimum-variance and mean-variance portfolios at every monthly rebalance date, using $\gamma = 158.10$ throughout.

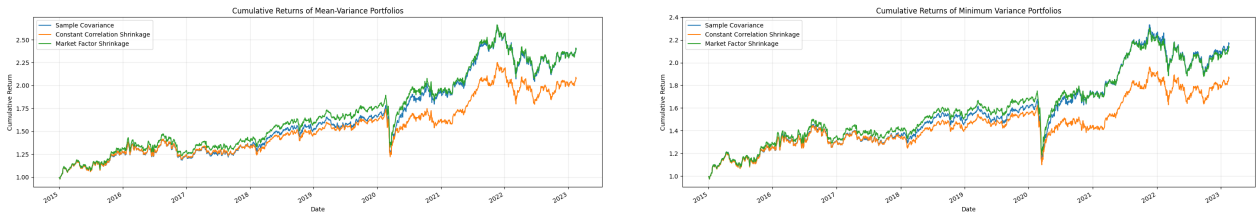


Figure 1: Cumulative returns for the Mean-Variance and Minimum-Variance portfolio.

Both Sample Covariance and Market Factor end at a cumulative return of 2.40, while Constant Correlation reaches 2.08. The sample covariance is known to amplify estimation noise, but its aggressive bets can sometimes pay off over specific periods. Shrinkage methods trade some return for stability: they impose structure on the covariance matrix, which reduces estimation error and typically delivers lower out-of-sample variance.

Task 1e: Comparative Discussion

The standard optimizer often makes mistakes by trusting noisy data too much. Shrinkage fixes this by mixing that noisy data with a safer, simpler 'target' matrix. The two targets work differently: the Market Factor model links all asset returns to one main market index, while Constant Correlation simply assumes every pair of assets has the exact same average correlation.

1. Matrix Conditioning and Shrinkage Intensities

Condition Number: The Sample matrix has the highest condition number (~ 270); both shrinkage methods safely reduce this to 173 (Const. Corr.) and 206 (Mkt. Factor), stabilizing the matrix inversion.

Shrinkage Intensity (δ^*): The Market Factor requires a lower average correction (16.21% vs 18.91%), meaning its economic index-driven logic fits the underlying data more naturally than a purely statistical average.

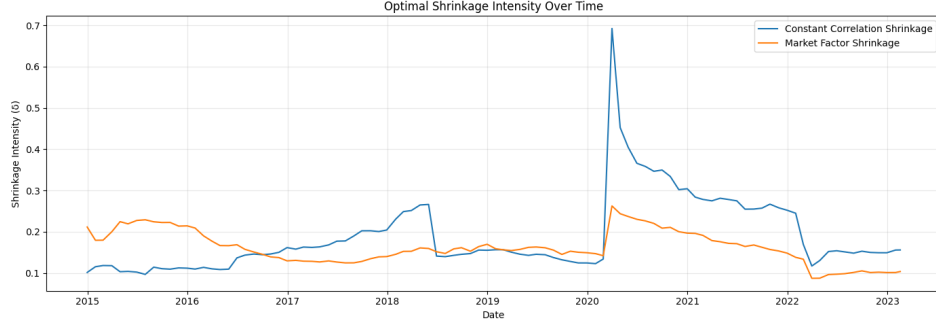


Figure 2: Evolution of Shrinkage Intensity (δ^*). Spikes highlight high-volatility market regimes where noise dominates and heavier shrinkage is required.

2. Out-of-Sample Portfolio Characteristics

Gross Exposure: Drops from 3.22 (Sample) to roughly 2.8 for both methods, significantly reducing leverage and extreme long-short positions.

Turnover: Falls from 65% to 51–54%, drastically cutting transaction costs which is crucial to protect the real Net Present Value (NPV) of the strategy.

Volatility: Remains stable at roughly 15.1% across the board, proving that the risk profile is preserved despite the drop in gross exposure.

Performance: The Market Factor matches the exact Cumulative Performance of the sample matrix (2.40), but achieves it with vastly superior operational efficiency.

Panel A: Covariance Estimator Properties

Metric	Sample Cov	Const. Corr	Market Factor
Avg. Shrinkage Intensity (δ^*)	0.00%	18.91%	16.21%
Avg. Condition Number	~ 270	~ 173	~ 206

Panel B: Out-of-Sample Portfolio Characteristics

Metric	Minimum Variance			Mean-Variance		
	Sample Cov	Const. Corr	Mkt Factor	Sample Cov	Const. Corr	Mkt Factor
Avg. Gross Exposure	3.14	2.70	2.81	3.22	2.77	2.88
Avg. Turnover	53%	41%	43%	65%	51%	54%
Avg. Rolling Annualized Vol.	14.53%	14.32%	14.32%	15.30%	15.11%	15.14%
Cumulative Performance	2.17	1.86	2.14	2.40	2.08	2.40

Table 2: Out-of-sample performance and risk characteristics of Minimum Variance and Mean-Variance portfolios.

Part II: Statistical Factor Model

Task 2a: PCA Implementation

We perform eigendecomposition of the sample covariance matrix at each rebalance date using `numpy.linalg.eigh`. Eigenvectors are aligned across consecutive periods via sign correction: if $u_k^{(t)} \cdot u_k^{(t-1)} < 0$, the sign of $u_k^{(t)}$ is flipped.

Task 2b: Results Discussion

Cosine similarity. PC1 (the market factor) maintains near-perfect stability ($\cos \approx 1.0$) across the entire sample. PC2 is also largely stable, with occasional drops. PC3 and PC4 show progressively

lower and more volatile cosine similarity, with frequent drops below 0.9. This confirms that higher-order eigenvectors are significantly unstable — a key limitation of statistical factor models.

The instability worsens during regime changes (2015, 2020), where the covariance structure shifts rapidly and even PC2 can rotate significantly. A shorter estimation window would amplify this effect, as less data means noisier eigenvector estimates.

Variance explained. On average, approximately 15–20% of the principal components are needed to explain 75% of total variance. During high-volatility periods (e.g., COVID-19), PC1 alone explains a larger fraction of variance because correlations spike and the market factor dominates. In calmer periods, the variance is more evenly distributed across components.

Covariance vs. correlation PCA. We compare the proportion of PCs needed to explain 75% of total variance under both approaches. PCA on the correlation matrix consistently requires more components ($\sim 20\text{--}35\%$) than PCA on the covariance matrix ($\sim 15\text{--}33\%$). The reason is that the correlation matrix treats all assets equally: no single volatile stock can dominate PC1, so variance spreads more evenly across components. Both methods show the same trend over time. During market crises (like 2016 or 2020), all stocks move together, so we need fewer PCs to reach the 75% target. In calm periods, stocks move independently, so we need more PCs. Since all Euro Stoxx 50 constituents are equities with broadly similar volatility levels, PCA on the covariance matrix is the more natural choice.

Task 2c: Factor Interpretation (January 31, 2019)

We plot the eigenvector loadings of the first three PCs, colored by GICS sector:

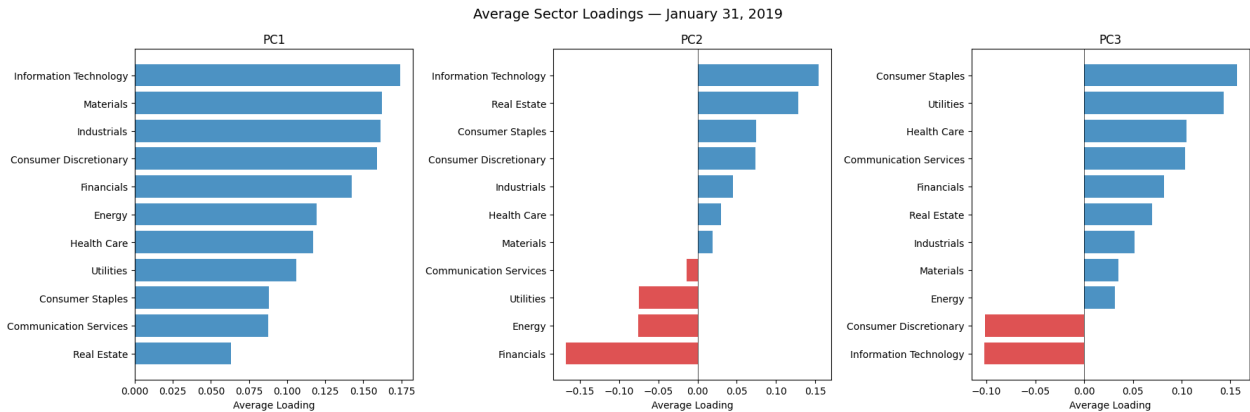


Figure 3: Factor loadings of the first three principal components at the reference date (January 31, 2019), colored by GICS sector.

PC1 — Market factor. All sectors load positively, so PC1 captures the overall market direction. Information Technology and Materials have the largest weights; Real Estate and Communication Services the smallest.

PC2 — Financials vs. Growth. Financials have a large negative loading, followed by Energy and Utilities. On the positive side we find Information Technology, Real Estate, and Consumer Staples. This factor separates rate-sensitive sectors from growth and defensive ones.

PC3 — Defensives vs. Cyclical. Consumer Staples, Utilities, and Health Care are positive; Information Technology and Consumer Discretionary are negative. This factor distinguishes sectors that hold up during slowdowns from those that depend on economic growth.

Part III: Connecting the Dots

Task 3a: Behavior of the eigenvalue spectra

In this section, we analyze the eigenvalue spectra on January 31, 2019. We compare four estimators: the Sample Covariance, the Constant Correlation Shrinkage, the Market Factor Shrinkage, and the PCA Denoised Matrix.

The **Sample Covariance** overestimates large eigenvalues (risk) and underestimates small eigenvalues (noise). The other methods fix this bias in two different ways.

PCA Denoised Matrix:

We compute this using our `pca_denoise_covariance` function. This method applies a “hard” cut. It keeps the largest k eigenvalues exactly the same (the true market signal). Then, it replaces all the remaining eigenvalues with their average value.

$$\hat{\Sigma}_{\text{denoised}} = \sum_{i=1}^k \lambda_i v_i v_i^\top + \bar{\lambda}_{\text{noise}} \sum_{i=k+1}^n v_i v_i^\top$$

On the graph, the red line follows the Sample line at first, then becomes a perfectly flat horizontal line (the noise floor). Naturally, as we increase k through the assigned values $\{1, 3, 5, 10\}$, this cutoff point shifts to the right, and our denoised matrix gradually converges back to the noisy Sample Covariance.

Shrinkage Estimators:

The two shrinkage estimators apply a “soft” correction. Instead of a hard cut, they gently pull all eigenvalues toward a structured target. They smoothly reduce large eigenvalues and increase small ones. We do not need to choose a cutoff parameter; the formula automatically calculates the optimal intensity δ^* from the data.

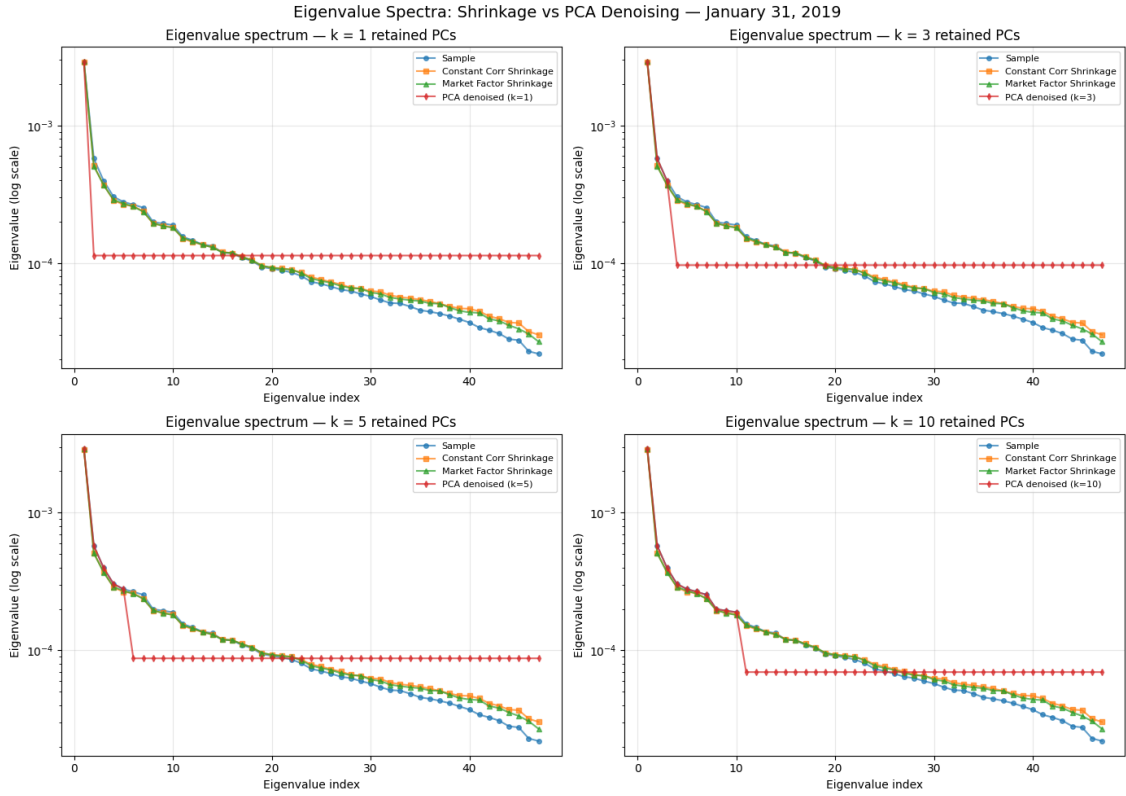


Figure 4: Eigenvalue Spectra: Shrinkage vs PCA Denoising on January 31, 2019.

Task 3b: Limited cross-sectional dimension

Looking at the graph, the Shrinkage lines almost perfectly overlap with the Sample Covariance line. This happens because of our dataset size. We are using the Euro Stoxx 50 index, which has only $n = 50$ assets, and our rolling window uses about $T = 504$ daily observations. This means we have 10 times more data points than assets ($T/n \approx 10$).

Because of this high ratio, the Sample Covariance is already stable, well-conditioned, and easy to invert. The estimation error (π) is low. Therefore, the optimal shrinkage intensity δ^* is relatively small (exactly 0.1556 for Constant Correlation and 0.1610 for Market Factor). Since the Sample Covariance still keeps about 84% of the total weight, the final matrices are very similar. This explains why their lines overlap on the graph.

Shrinkage shows its massive benefits only when we analyze a lot of assets, specifically when the number of assets is close to or larger than the observations ($n \geq T$). In that scenario, the Sample Covariance breaks down, and Shrinkage provides a vital correction. For a small universe like the Euro Stoxx 50, the Sample Covariance is already reliable, so the Shrinkage methods correctly apply a small correction.

Identified Errors

Errors in shrinkage.py — constant_corr_shrinkage

- `std_devs = variances`: missing `np.sqrt()`. Corrected to `std_devs = np.sqrt(variances)`.
- `diff = outer_prod + S`: sign error. Corrected to `outer_prod - S`.
- `2 * avg_corr`: incorrect coefficient in $\hat{\rho}$ off-diagonal term. Corrected to `avg_corr / 2`.
- `pi_hat /= T` and `rho_hat /= T` placed inside the loop: moved outside.
- Missing division by T in shrinkage intensity: `numerator / denominator` corrected to `numerator / (T * denominator)`.
- `shrunk_cov: cov_matrix + target`: incorrect formula. Corrected to `intensity * target + (1 - intensity) * cov_matrix`.

Errors in shrinkage.py — market_factor_shrinkage

- `market_variance = market_aligned.std()`: should be `.var()`.
- `diff = outer_prod + S`: same sign error as above. Corrected to `outer_prod - S`.
- `pi_hat = np.sum(diff**2)`: overwrites instead of accumulating. Corrected to `+=`.
- Missing `pi_hat /= T` after the loop: `rho_hat` is correctly divided by T , but `pi_hat` is not.
- Shrunk covariance has δ and $(1-\delta)$ swapped: `intensity * cov_matrix + (1 - intensity) * target` corrected to `intensity * target + (1 - intensity) * cov_matrix`.

Errors in backtest.py

- `.bfill()`: backward fill propagates future portfolio weights to past dates (look-ahead bias). Corrected to `.ffill()`.

Errors in ex3_notebook.ipynb

- `time_horizon = 3 * 252`: the assignment specifies 2 years. Corrected to `2 * 252`.